

7.11.2015(E)

K. J. Somaiya College of Engineering, Mumbai-77
(Autonomous College Affiliated to University of Mumbai)

End Semester Exam

Nov – Dec 2015

Max. Marks:

Class: TE

Name of the Course: Computer Networks

Course Code: UCEC504

Duration: 3hrs

Semester: V

Branch: Computer Engineering

Instructions:

- (1) All Questions are Compulsory
- (2) Draw neat diagrams
- (3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1 (a)	Explain in short different framing methods.	5M
Q 1 (b)	What are the design issues for the OSI layer model?	5M
Q1 (c)	List the advantages of fiber optics as a communication medium.	5M
Q1 (d)	Received codeword for a communication system is 11011100101. Identify error bit if any and write the correct codeword using Hamming code algorithm.	5M
Q2 (a)	Write about congestion control algorithms in detail.	10M
	OR	
Q2 (a)	State flow characteristics. Explain various techniques to improve QoS in detail.	2M 8M
Q.2 (b)	Explain IPv4 header format in detail with explanation for each field.	10M
Q3 (a)	Explain pure ALOHA in detail.	10M
	OR	
Q3 (a)	Calculate internet checksum for the following message at sender side and verify the checksum at receiver side: 466F726F757A616E	10M
Q3 (b)	Explain Sliding window protocol with Go-Back-N technique.	10M
	OR	
Q3 (b)	Explain Sliding window protocol with Selective Repeat ARQ technique.	10M

Q4 (a)	An organization is granted a block of addresses with the beginning address 14.24.74.0/24. There are $2^{32-24} = 256$ addresses in this block. The organization needs to have 11 subnets as shown below: (a) Two subnets, each with 64 addresses. (b) Two subnets, each with 32 addresses. (c) Three subnets, each with 16 addresses. (d) Four subnets, each with 4 addresses. Design the subnets and draw necessary diagram.	10M
Q4 (b)(1)	Draw the routing table for router R1 in the network shown below.	05M
Q4(b)(2)	Explain the next hop and route method for forwarding.	05M
Q5 (a)	Write Short Notes On : (any two) (a) Bluetooth (b) TELNET (c) SNMP	10M
Q5 (b)	Define Topology. Explain various types of network topologies in detail	10M